

New Lower Bounds for C_4 -Free Subgraphs of the Hypercubes Q_6 , Q_7 , and Q_8 : Constructions, Structure, and Computational Method

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Abstract

We establish new lower bounds $\text{ex}(Q_7, C_4) \geq 304$ and $\text{ex}(Q_8, C_4) \geq 680$ for the maximum number of edges in a C_4 -free subgraph of the 7- and 8-dimensional hypercubes, and give a modern computational reproduction of $\text{ex}(Q_6, C_4) = 132$. All bounds are witnessed by explicit constructions certified by exhaustive enumeration of all four-cycles (240 for Q_6 , 672 for Q_7 , 1792 for Q_8).

For Q_7 we identify 19 866 distinct C_4 -free subgraphs on 304 edges; their *dimension profiles* fall into exactly 20 types. All 19 866 solutions share a rigid structural core: degree sequence $\{4^{32}, 5^{96}\}$, spectral radius $\lambda_1 \approx 4.787$, and local maximality. Pairwise Hamming distances range from 36 to 260. Whether these solutions exhaust all 304-edge C_4 -free subgraphs of Q_7 remains open.

For Q_8 we analyse the local structure of the 680-edge construction: every non-edge of the construction creates at least one C_4 , and 1 076 independent searches at 681 edges did not achieve zero violations. These observations constitute computational evidence, not a proof, of the conjectured equality $\text{ex}(Q_8, C_4) = 680$.

The constructions are found by a two-phase simulated annealing algorithm with $\text{Aut}(Q_n)$ -based diversification, described in full detail. For Q_6 we additionally provide an ILP-based proof that $\text{ex}(Q_6, C_4) \leq 132$, reproducing the exact value computationally.

Edge lists, ILP files, and source code are publicly available at <https://github.com/minamominamoto/c4free-hypercube>.

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Keywords: hypercube, C_4 -free, extremal graph theory, simulated annealing, integer linear programming, dimension profile.

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1 Introduction

The n -dimensional hypercube Q_n has vertex set $\{0, 1\}^n$, two vertices adjacent iff they differ in exactly one coordinate; it has 2^n vertices and $n \cdot 2^{n-1}$ edges. The problem of determining $\text{ex}(Q_n, C_4) = \max\{|E(G)| : G \subseteq Q_n, G \text{ is } C_4\text{-free}\}$ was raised by Erdős [6] and remains open in general.

The best known asymptotic bounds are

$$\frac{1}{2}(n + \sqrt{n}) \cdot 2^{n-1} \leq \text{ex}(Q_n, C_4) \leq 0.6032 \cdot n \cdot 2^{n-1}, \quad (1)$$

where the lower bound (valid for $n = 4^r$) is due to Brass, Harborth, and Nienborg [3], and the upper bound to Baber [1] via the flag algebra method. Balogh et al. [2] give 0.6068 via a related method. For general $n \geq 9$, the weaker bound $\frac{1}{2}(n + 0.9\sqrt{n}) \cdot 2^{n-1}$ applies [3]. Exact values have been determined for small n ; Harborth and Nienborg [7] proved $\text{ex}(Q_6, C_4) = 132$.

Our main results are:

Theorem 1. (i) $\text{ex}(Q_7, C_4) \geq 304$.

(ii) $\text{ex}(Q_8, C_4) \geq 680$.

Both bounds are realised by explicit constructions (supplementary files `q7_edges_304.json1` and `q8_edges_680.json1`).

The paper is organised as follows. Section 2 describes the C_4 enumeration and verification procedure used throughout. Sections 3–4 present the Q_7 and Q_8 constructions. Section 5 classifies the 19 866 Q_7 solutions found by their dimension profiles, yielding exactly 20 types. Section 6 describes the two-phase simulated annealing algorithm. Section 7 reproduces the exact value $\text{ex}(Q_6, C_4) = 132$ via ILP. Section 8 lists open problems.

Table 1: Known values and new lower bounds of $\text{ex}(Q_n, C_4)$. The BHN column gives the Brass–Harborth–Nienborg general estimate.

n	$ E(Q_n) $	$\text{ex}(Q_n, C_4)$	BHN est.	Source
4	32	24	≈ 25.6	Chung [4]
5	80	56	≈ 51.2	Emamy-K et al. [5]
6	192	132	≈ 96.0	Harborth–Nienborg [7] (reproduced)
7	448	$\geq \mathbf{304}$	≈ 300.2	This work
8	1024	$\geq \mathbf{680}$	≈ 674.9	This work

2 C_4 Enumeration and Verification

Lemma 2 (C_4 enumeration in Q_n). *Every four-cycle of Q_n is uniquely determined by a pair of coordinate directions $0 \leq i < j \leq n - 1$ and a base vertex $b \in \{0, 1\}^n$ with $b_i = b_j = 0$. Its four vertices are*

$$b, \quad b \oplus e_i, \quad b \oplus e_j, \quad b \oplus e_i \oplus e_j,$$

and the total number of four-cycles in Q_n is $\binom{n}{2} \cdot 2^{n-2}$. For $n = 6$: 240; for $n = 7$: 672; for $n = 8$: 1792.

Verification algorithm. Given a candidate edge set $E \subseteq E(Q_n)$ stored as a hash set, the following procedure performs a complete, deterministic C_4 -freeness check:

```

for each pair  $(i, j)$  with  $0 \leq i < j \leq n - 1$ :
  for each  $b \in \{0, 1\}^n$  with  $b_i = b_j = 0$ :
    let  $v_1 = b, v_2 = b \oplus e_i, v_3 = b \oplus e_j, v_4 = b \oplus e_i \oplus e_j$ 
    if  $\{v_1 v_2, v_1 v_3, v_2 v_4, v_3 v_4\} \subseteq E$ : return VIOLATION
  return C4-FREE

```

We applied this algorithm to all constructions and all 19 866 solutions of Q_7 ; every call returned C4-FREE.

3 The Q_7 Construction

We constructed a C_4 -free subgraph of Q_7 on 304 edges by penalty-based simulated annealing (Section 6), then certified it by Lemma 2’s algorithm.

Proposition 3. *The 304-edge construction has the following properties:*

- (i) *Degree sequence: 32 vertices of degree 4 and 96 of degree 5 (degree sum $608 = 2 \cdot 304$).*
- (ii) *Bipartite: $\lambda_1 \approx 4.787, \lambda_n \approx -4.787, \lambda_1 + \lambda_n = 0$ to machine precision.*
- (iii) *$\text{Tr}(A^4) = 5216$, attaining the C_4 -free trace equality $\sum_i \deg(i)^2 + \sum_{uv \in E} (\deg u + \deg v) - 2|E|$.*
- (iv) *Local maximality: every non-edge of Q_7 creates a C_4 when added.*

Through 19 866 independent runs (Section 5.3) we found 19 866 C_4 -free subgraphs of Q_7 on 304 edges with pairwise distinct edge sets (not quotiented by $\text{Aut}(Q_7)$), all sharing the structural properties stated in Proposition 9.

4 The Q_8 Construction

Simulated annealing produced a C_4 -free subgraph of Q_8 on 680 edges, certified by exhaustive enumeration of all 1 792 four-cycles.

Proposition 4. *The 680-edge C_4 -free subgraph of Q_8 has the following properties:*

- (i) *Degree sequence $\{4^8, 5^{160}, 6^{88}\}$; degree sum $1360 = 2 \cdot 680$.*
- (ii) *Bipartite, with $\lambda_1 + \lambda_n = 0$.*
- (iii) *Local maximality: every non-edge of Q_8 creates a C_4 upon addition.*
- (iv) *Edge density $680/1024 = 66.4\%$, compared to $304/448 = 67.9\%$ for the Q_7 construction.*

Remark 5. Since 8 is not a power of 4, the exact BHN formula does not apply for $n = 8$. The applicable estimate is $\frac{1}{2}(8 + 0.9\sqrt{8}) \cdot 128 \approx 674.9$, which our 680-edge construction surpasses by 5.1 edges.

4.1 Local optimality of the 680-edge construction

For the 680-edge solution, every non-edge of Q_8 creates at least one C_4 upon addition. Among the 344 non-edges: exactly 1 creates exactly 1 new C_4 ; exactly 1 creates exactly 2; the remaining 342 create 3 or more.

Over 1 076 independent simulated-annealing trials targeting 681 edges, the minimum C_4 violation count observed was 1 (never 0), where the violation count denotes the number of four-cycles fully contained in the candidate edge set.

Conjecture 6. $\text{ex}(Q_8, C_4) = 680$.

Table 2 compares the Q_7 and Q_8 constructions.

Table 2: Structural comparison of the Q_7 and Q_8 constructions.

Parameter	Q_7 (304 edges)	Q_8 (680 edges)
Total edges $ E(Q_n) $	448	1024
Achieved edges	304	680
Edge density	67.9%	66.4%
Min degree	4	4
Max degree	5	6
Degree sequence	$\{4^{32}, 5^{96}\}$	$\{4^8, 5^{160}, 6^{88}\}$
BHN estimate	≈ 300.2	≈ 674.9
Improvement	+3.8	+5.1

5 Structural Classification of Q_7 Solutions

5.1 Dimension profiles and the twenty types

Definition 7 (Dimension profile). For $G \subseteq Q_7$, let $e_i(G) = |\{uv \in E(G) : u \oplus v = 2^i\}|$. The *dimension profile* is the sorted tuple $\pi(G) = (e_{\sigma(0)} \geq \cdots \geq e_{\sigma(6)})$.

The dimension profile is invariant under coordinate permutations (a subgroup of $\text{Aut}(Q_7)$ of order $7! = 5040$), giving a coarse but computable classification.

Proposition 8. *Among the 19 866 solutions, the dimension profile takes exactly **20** distinct values.*

Table 3 lists all 20 types with frequencies and structural invariants.

Table 3: The 20 dimension-profile types of 304-edge C_4 -free subgraphs of Q_7 , sorted by decreasing frequency. H : Shannon entropy of normalised profile (bits).

Rank	Profile π	Solutions	H	λ_1 (mean)
1	[44, 44, 44, 44, 43, 43, 42]	3155	2.8072	4.787
2	[45, 45, 45, 43, 43, 42, 41]	2913	2.8065	4.788
3	[45, 45, 45, 43, 43, 43, 40]	2200	2.8063	4.786
4	[44, 44, 44, 44, 44, 43, 41]	2116	2.8069	4.787
5	[46, 46, 46, 42, 42, 42, 40]	1756	2.8053	4.788
6	[46, 46, 46, 42, 42, 41, 41]	1181	2.8054	4.789
7	[44, 44, 44, 44, 44, 44, 40]	1085	2.8066	4.787
8	[45, 45, 45, 43, 42, 42, 42]	1064	2.8066	4.787
9	[45, 45, 44, 43, 43, 43, 41]	974	2.8067	4.787
10	[44, 44, 44, 44, 44, 42, 42]	931	2.8070	4.787
11	[47, 47, 47, 41, 41, 41, 40]	902	2.8037	4.789
12	[45, 45, 43, 43, 43, 43, 42]	439	2.8069	4.787
13	[45, 44, 44, 43, 43, 43, 42]	307	2.8070	4.787
14	[46, 45, 45, 42, 42, 42, 42]	236	2.8063	4.787
15	[44, 44, 44, 43, 43, 43, 43]	163	2.8073	4.787
16	[47, 47, 44, 43, 41, 41, 41]	153	2.8050	4.788
17	[46, 46, 44, 42, 42, 42, 42]	107	2.8062	4.787
18	[48, 48, 48, 40, 40, 40, 40]	101	2.8014	4.789
19	[46, 46, 43, 43, 42, 42, 42]	44	2.8063	4.787
20	[46, 46, 45, 42, 42, 42, 41]	39	2.8058	4.787
Total		19866		

Observations: (1) λ_1 lies in $[4.786, 4.789]$ across all 20 types. (2) H ranges from 2.801 to 2.807, all near $\log_2 7 \approx 2.807$. (3) Type 18 ($[48, 48, 48, 40, 40, 40, 40]$) has 3-fold symmetry.

5.2 Shared structural properties

Proposition 9. *Every solution in the 19 866 satisfies:*

- (i) Degree sequence $\{4^{32}, 5^{96}\}$.
- (ii) $\lambda_1 + \lambda_n = 0$ (bipartite), $\lambda_1 \approx 4.787$.

- (iii) $\text{Tr}(A^4) = 5216$ (C_4 -free trace equality).
- (iv) *Local maximality: no edge of Q_7 can be added without creating a C_4 .*
- (v) *Every edge of Q_7 appears in at least one solution (verified by taking the union of the edge sets over all 19 866 solutions).*

5.3 Solution landscape

Pairwise Hamming distances $d_H(G, G') = |E(G) \Delta E(G')|$ over a random sample of 5 000 pairs range from 36 to 260 (mean ≈ 195.4 , median 196).

Over 1 076 independent trials targeting 305 edges, the minimum C_4 violation count was 2 (never 0).

Conjecture 10. $\text{ex}(Q_7, C_4) = 304$.

6 Computational Method

6.1 Two-phase simulated annealing

Phase 1 (Penalty SA). We minimise $f(E) = -|E| + \lambda V$ where V is the number of C_4 violations and $\lambda \in [0.30, 0.90]$. Temperature schedule: geometric from $T_0 \in [0.20, 4.00]$ to $T_1 \in [0.001, 0.030]$ over 3–15 million steps. A move toggles a single edge; the change in f is computed incrementally in $O(\deg)$ time using precomputed C_4 -to-edge incidence lists.

Phase 2 (Swap SA). Edge count is fixed; swap moves (remove one edge, add one non-edge) minimise V . Parameters: up to 2×10^8 steps for Q_7 at 305 edges; 5×10^7 for Q_8 at 681 edges.

Diversification. Before each trial, a random element of $\text{Aut}(Q_n)$ (random bit permutation composed with random bit flip) is applied to the current best solution. This prevents repeated exploration of the same automorphism class.

6.2 Comparison with known lower bounds

$$\begin{aligned} n = 7 : \quad & \frac{1}{2}(7 + 0.9\sqrt{7}) \cdot 64 \approx 300.2, \quad \text{our bound: } 304 \ (\Delta = +3.8, +1.25\%), \\ n = 8 : \quad & \frac{1}{2}(8 + 0.9\sqrt{8}) \cdot 128 \approx 674.9, \quad \text{our bound: } 680 \ (\Delta = +5.1, +0.75\%). \end{aligned}$$

6.3 Regularity trend across dimensions

Degree sequences across dimensions suggest increasing regularity: Q_6 : $\{4^{32}, 5^{32}\}$; Q_7 : $\{4^{32}, 5^{96}\}$; Q_8 : $\{4^8, 5^{160}, 6^{88}\}$. The degree support $\{n-3, n-2, n-1\}$ may reflect an underlying structural principle for optimal C_4 -free subgraphs of Q_n .

7 Computational Proof that $\text{ex}(Q_6, C_4) = 132$

Harborth and Nienborg [7] proved $\text{ex}(Q_6, C_4) = 132$ by combinatorial arguments. We reproduce this result computationally, providing an independent verification and a template for small hypercubes.

7.1 Lower bound: explicit construction

Simulated annealing (Section 6) finds a C_4 -free subgraph of Q_6 on 132 edges, certified by exhaustive enumeration of all 240 four-cycles. The explicit edge list is provided in `q6_edges_132.jsonl`.

7.2 Upper bound: integer linear program

Label the 192 edges of Q_6 as $x_1, \dots, x_{192} \in \{0, 1\}$. For each of the 240 four-cycles $\{e_1, e_2, e_3, e_4\}$, the constraint $x_{e_1} + x_{e_2} + x_{e_3} + x_{e_4} \leq 3$ ensures C_4 -freeness. The ILP is:

$$\max \sum_{i=1}^{192} x_i \quad \text{subject to} \quad \sum_{e \in C} x_e \leq 3 \quad \forall C \in \mathcal{C}_6, \quad x_i \in \{0, 1\}.$$

This formulation has 192 binary variables and 240 constraints. The MPS file `q6_ilp.mps` is provided for independent verification.

Proposition 11. *The optimal value of the above ILP is 132, establishing $\text{ex}(Q_6, C_4) \leq 132$. Combined with the lower bound construction, $\text{ex}(Q_6, C_4) = 132$.*

On standard desktop hardware, standard MIP solvers such as SCIP, HiGHS, or Gurobi verify this instance quickly.

8 Open Problems

1. Prove or disprove $\text{ex}(Q_7, C_4) = 304$ (Conjecture 10).
2. Determine $\text{ex}(Q_8, C_4)$ exactly (Conjecture 6).
3. What is the number of $\text{Aut}(Q_7)$ -orbits among all 304-edge C_4 -free subgraphs of Q_7 ?
4. Is the degree sequence $\{4^{32}, 5^{96}\}$ necessary for any locally maximal C_4 -free subgraph of Q_7 on 304 edges?
5. Can the flag algebra method of [1, 2] yield tight bounds for small $n \leq 8$?
6. Does the degree sequence pattern $\{(n-3)^a, (n-2)^b, (n-1)^c\}$ continue for $n = 9, 10, \dots$?

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References

- [1] R. Baber, Turán densities of hypercubes, arXiv:1201.3587, 2012.
- [2] J. Balogh, P. Hu, B. Lidický, and H. Liu, Upper bounds on the size of 4- and 6-cycle-free subgraphs of the hypercube, *Eur. J. Combin.* **35** (2014), 75–85.
- [3] P. Brass, H. Harborth, and H. Nienborg, On the maximum number of edges in a C_4 -free subgraph of Q_n , *J. Graph Theory* **19** (1995), 17–23.
- [4] F. Chung, Subgraphs of a hypercube containing no small even cycles, *J. Graph Theory* **16** (1992), 273–286.
- [5] M.R. Emamy-K., P. Guan, and P.I. Rivera-Vega, On the characterization of the maximum squareless subgraphs of the 5-cube, *Congr. Numer.* **88** (1992), 97–109.
- [6] P. Erdős, On some problems in graph theory, combinatorial analysis and combinatorial number theory, in: *Graph Theory and Combinatorics* (B. Bollobás, ed.), Academic Press, 1984, pp. 1–17.
- [7] H. Harborth and H. Nienborg, Maximum number of edges in a six-cube without four-cycles, *Bull. ICA* **12** (1994), 55–60.
- [8] A. Thomason and P. Wagner, Bounding the size of square-free subgraphs of the hypercube, *Discrete Math.* **309** (2009), 1730–1735.